(d) Interference

Ans. (a) : Optical fibre works on the principle of total internal reflection.
Total internal reflection is a phenomenon in which the angle of incidence is slightly higher than the critical angle and the light does not pass through the rarer medium but entire lights is reflected back to the denser medium.
For total internal reflection of light, it is necessary that light is passing from denser medium to rarer medium or the angle of incidence is greater than the critical angle.

92. Suppose a rocket ship is receding from the earth at a speed of 2/10th the velocity of light. A light in the rocket ship appears blue to the passengers on the ship. What colour would it appear to an observer on the earth?
(a) Blue (b) Orange
(c) Yellow (d) Yellow-orange

Ans. (c) : As the portion of light source reflected back from the earth, its frequency decreases and thus wavelength increases as a result of which spectrum line is shifted toward the red of spectrum.

$$\text{Wavelength change} - \Delta\lambda = \frac{Vs \times d}{C}$$

$$= \frac{2}{10} \quad \left[\begin{array}{l} \Delta\lambda = \frac{v}{f} \\ f = \frac{c}{\lambda} \end{array}\right.$$

$$= \frac{2}{10} \times 4700°A \quad \left[\begin{array}{l} \lambda \text{ of blue light} \\ = 4700°A \end{array}\right.$$

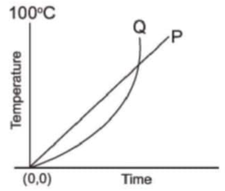
$$= 940°A$$

Wavelength visible to earth observe
(λ visible) = 47000 + 940 = 5640°A
λ yellow light $\simeq$ 5640°A
So, it falls in range of yellow spectrum.

93. Cryogenic engines find applications in:
(a) Sub-marine propulsion
(b) Frost-free refrigerators
(c) Rocket technology
(d) Researches in superconductivity

Ans. (c) : Cryogenic engine finds most of its application in rocket technology.
A cryogenic engine is a rocket engine that uses a cryogenic fuel or oxidizer that is, its fuel or oxidizer (or both) are gases liquefied and stored at very low temperature.
A cryogenic rocket stage is more efficient and provides more thrust for every kilogram of propellant it burns compared to solid and earth-storable liquid propellant rocket stages. Specific impulse achievable with cryogenic propellants is much higher compared to earth storable liquid and solid propellants, giving it a substantial payload advantage.

94. The variations in temperatures from 0°C to 100°C with respect to time of two liquids P, Q are shown in the graph given below. Which one of the following statements is correct?



(a) During heating, liquid P remained hotter than liquid Q throughout
(b) (b) At no point of time during heating did the two liquids have the same temperature
(c) P attained the temperature of 100°C faster than Q
(d) Q attained the temperature of 100°C faster than P

Ans. (d) : From graph it is clear that Q attained the temperature of 100°C faster than P.

95. An air bubble in water will act like a:
(a) Convex mirror (b) Convex lens
(c) Concave mirror (d) Concave lens

Ans. (d) : When lens is immersed in a medium whose refractive index is greater than that of the material of the lens, the nature of lens is reversed and thus convex lens behave like concave ad vice-versa. The air bubble is convex lens, but when it is submersed into water, which is having higher refractive index than air and thus lens behave like concave lens.

CHEMISTRY

96. Which one of the following is a mixed fertilizer?
(a) CAM (b) Urea
(c) Ammonium Sulphate (d) NPK

Ans. (c) : Mixed fertilizers are fertilizers which contains two or more of the elements of nitrogen, phosphorus and potassium, which are essential for promoting plant growth and high crop yields. They are obtained by through mixing the ingredients either manually or mechanically Ammonium sulfate is inorganic salt and is used as soil fertilizers. It contains 21% Nitrogen and 24% sulphur.

97. Which one of the following polymers is widely used for making bullet proof material?
(a) Polyvinyl chloride (b) Polyamides
(c) Polyethylene (d) Polycarbonates

Ans. (d) : Polycarbonates are widely used for making bullet proof material.
Polycarbonates are extremely durable and has high impact resistance, event against violent blows. Bullet resistant polycarbonate is more suitable when protecting against ballistics, including heavy automatic fire armor.

Polyvinyl chloride are used for making sheet covers, sheets, film etc.
Polyethylene are used for making plastic covers, plastic bottle, pipes and packaging covers.

98. The chemical used as a 'fixer' in photography is:
(a) Sodium sulfate
(b) Sodium thiosulfate
(c) Ammonium persulfate
(d) Borax

Ans. (b) : The chemical used as a fixer in photography is sodium thiosulfate. It is also known as sodium hyposulfite and is used by photographers to fix developed negatives and prints. It acts by dissolving the part of the silver salts coated onto film which remain unchanged by exposure to light.

99. Which one of the following can be used to confirm whether drinking water contains a gamma emitting isotope or not?
(a) Microscope
(b) Lead plate
(c) Scintillation counter
(d) Spectrophotometer

Ans. (c) : The presence of gamma emitting isotopes in drinking water is detected by the scintillation counter.

100. Which one of the following pairs of materials serves as electrodes in rechargeable batteries commonly used in devices such as torch lights, electric shavers, etc?
(a) Nickel and cadmium
(b) Zinc and carbon
(c) Lead peroxide and lead
(d) Iron and cadmium

Ans. (a) : Nickel and Cadmium rod act as electrodes in the rechargeable batteries commonly used in devices such as torch light, electric shavers etc.
Nickel rod act as cathode and cadium rod act as anode and they react in presence of electrolyte solution of Potassium Hydro oxide in Ni-Cl batteries.

101. The difference between a nuclear reactor and an atomic bomb is that:
(a) No chain reaction takes place in nuclear reactor while in the atomic bomb there is a chain reaction
(b) The chain reaction in nuclear reactor is controlled
(c) The chain reaction in nuclear reactor is not controlled
(d) No chain reaction takes place in atomic bomb while it takes place in nuclear reactor

Ans. (b) : In nuclear reactors, the reaction is moderated and controlled for use of peaceful and constructive purpose. Where as in nuclear bomb uncontrolled chain reaction takes place and it is used for destructive purposes. In nuclear bomb high amount of energy is released as compared to nuclear reactor. Nuclear reactors uses cadmium rod as moderator and water and graphite as coolant.

BIOLOGY

102. The normal temperature of human body on the Kelvin scale is:
(a) 280 (b) 290
(c) 300 (d) 310

Ans. (d) : The normal temperature of human body is 37°C
To convert °C into Kelvin
= [273 + °C] Kelvin
= 273 + 37
= 310K

103. Which one of the following sets is correctly matched?
(a) Diphtheria, Pneumonia and Leprosy: Hereditary
(b) AIDS, Syphilis and Gonorrhoea : Bacterial
(c) Colour blindness, Hemophilia and Sickle cell anaemia : Sex linked
(d) Polio, Japanese B encephalitis and plague: Viral

Ans. (c) : Syphilis, Gonorhoea – Sexually transmited diseases Diphteria Leprosy and Plague – bacterial diseases AIDS, Polio, Japanese B encephalitis – Viral colour blindness, Hemophillia and sickle cell anaemia – sex linked.
Thus only option (c) is matched correctly.

104. A person with 'AB' blood group is sometimes called a universal recipient because of the:
(a) Lack of antigen in his blood.
(b) Lack of antibodies in his blood.
(c) Lack of both antigens and antibodies in his blood.
(d) Presence of antibodies in his blood.

Ans. (b) : A person with 'AB' blood group is sometimes called universal recipient because it have antigens of 'A' and 'B' blood group whereas it does not have any antibodies in his body. So 'AB' blood group can receive bloods from person of 'A', 'B', 'AB' and 'O' blood groups.

105. Which one of the following is a membrane that protects the developing embryo from desiccation?
(a) Amnion (b) Allantois
(c) Chorion (d) Yolk sac

Ans. (a) : The amnion is a thin, tough membrane that protects a developing child. It allows for nutrients to reach the fetus and wastes to be removed. Amniotic fluid is found inside amnion which will protect the developing embryo from desiccation.

106. Which one of the following antimicrobial drugs is suitable for treatment of both tuberculosis and leprosy?
(a) Isoniazid
(b) p-aminosalicylic acid